

Review article

Review of bidirectional DC–DC converter topologies for hybrid energy storage system of new energy vehicles



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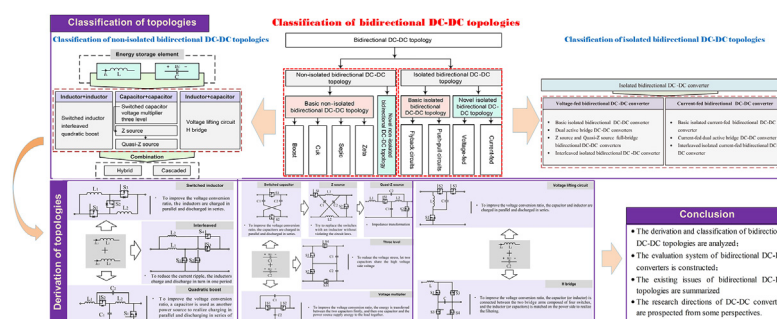
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HIGHLIGHTS

- The derivation and classification of bidirectional DC–DC topologies are analyzed.
- The evaluation system of bidirectional DC–DC converters is constructed.
- The existing issues of bidirectional DC–DC topologies are summarized.
- The research directions of DC–DC converters are prospected from some perspectives.

GRAPHICAL ABSTRACT



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ABSTRACT

New energy vehicles play a positive role in reducing carbon emissions. To improve the dynamic performance and durability of vehicle powertrain, the hybrid energy storage system of “fuel cell/power battery plus super capacitor” is more used in new energy vehicles. Bidirectional DC–DC converters with wide voltage conversion range are essential for voltage matching and power decoupling between super capacitor and vehicle bus, helping to improve the low input voltage characteristics of super capacitors and realize the recovery of feedback energy. In recent years, the topologies of bidirectional converters have been widely investigated and optimized. Aiming to obtain bidirectional DC–DC converters with wide voltage conversion range suitable for hybrid energy storage system, a review of the research status of non-isolated converters based on impedance networks and isolated converters based on transformer are presented. Additionally, an evaluation system for bidirectional DC–DC topologies for hybrid energy storage system is constructed, providing a reference for designing bidirectional DC–DC converters. The performance of eight typical non-isolated converters and seven typical isolated converters are comprehensively evaluated by using this evaluation system. On this basis, issues about DC–DC converters for hybrid energy storage system are discussed, and some suggestions for the future research directions of DC–DC converters are proposed. The optimization of bidirectional DC–DC converters for hybrid energy storage system from the perspectives of wide bandgap device application, electromagnetic compatibility technology and converter fault diagnosis strategies is the main research direction.

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1. Introduction

The consumption of fossil fuels such as oil and natural gas, and the increase of greenhouse gas emissions are having a serious impact on the climate and environment. With the increasing shortage of non-renewable energy, energy security has gradually been paid attention by governments. In recent years, vehicle ownership in various countries has increased year by year, and new energy vehicles (NEV) with unconventional vehicle fuels as power sources help to reduce the dependence of transport on non-renewable energy and reduce pollution emissions, and their ratio in vehicles is increasing year by year. According to the consultative draft of New Energy Automobile Industry Development Plan (2021–2035), China's market share of NEV will be 20% by 2025 [1]. Hydrogen fuel cell vehicle (FCV) holdings are expected to reach 100,000

in 2025 and 1 million in 2035, according to Technology Roadmap for Energy Saving and New Energy Vehicles 2.0 [2]. Plug-in hybrid electric vehicle (PHEV) holdings are expected to peak in 2025 [3]. In addition, BloombergNEF predicts that by the end of 2022, more than 26 million NEV will be on the road around the world [4].

FCV, PHEV and plug-in fuel cell vehicle (FC-PHEV) are the typical NEV. The hybrid energy storage system (HESS) is general used to meet the requirements of power density and energy density of NEV [5]. The structures of HESS for NEV are shown in Fig. 1. HESS for FCV is shown in Fig. 1(a) [6]. Fuel cell (FC) provides average power and the super capacitor (SC) provides the instantaneous power when the operating conditions change instantaneously [7]. The unidirectional DC–DC converter realizes voltage matching and power decoupling between FC and DC bus [8]. The bidirectional DC–DC converter realizes voltage matching between SC and DC

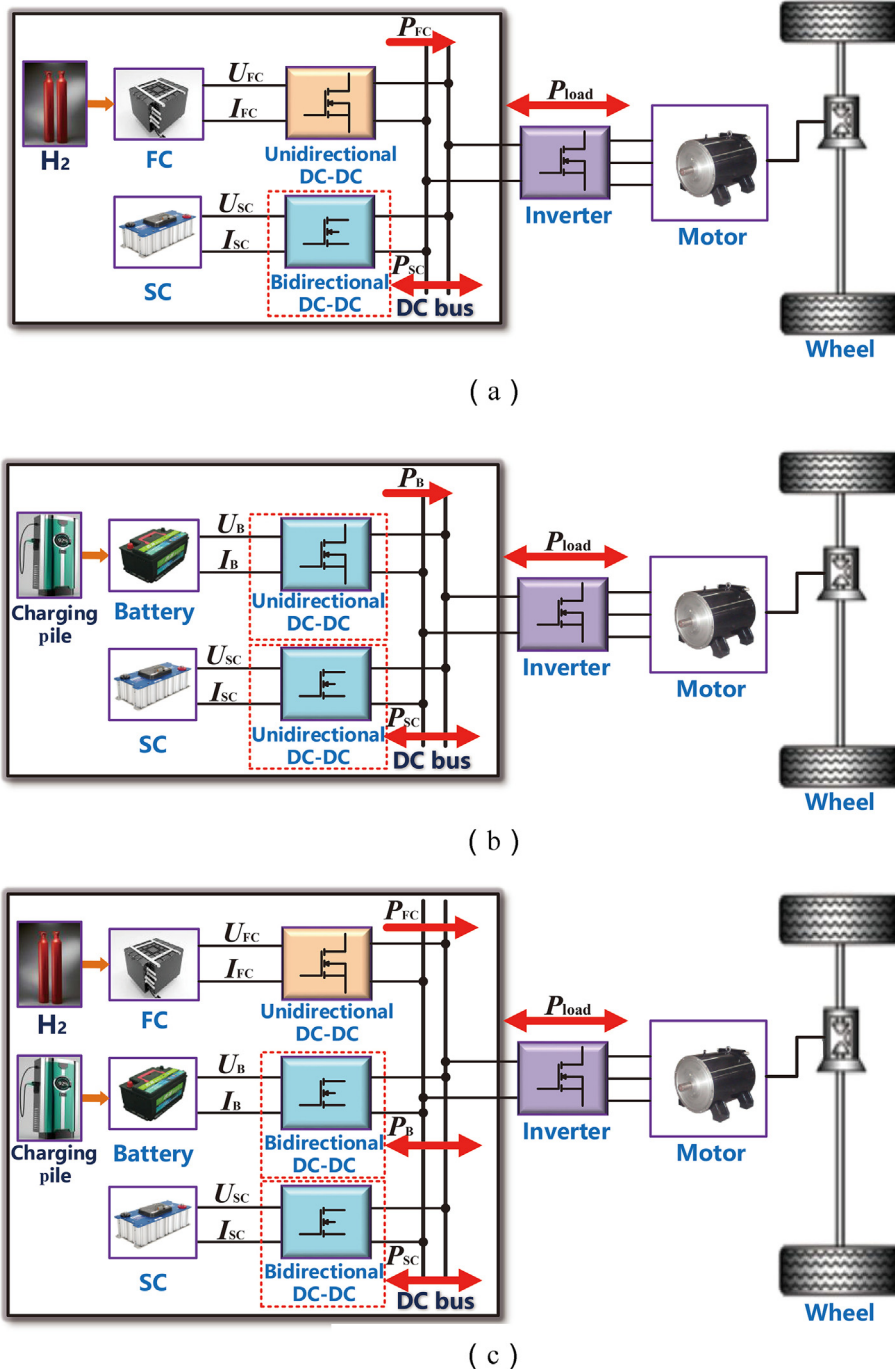


Fig. 1. Structure diagrams of HESS for NEV. (a) HESS for FCV. (b) HESS for PHEV. (c) HESS for FC-PHEV.

bus, and bidirectional flow of instantaneous energy. HESS for PHEV is shown in Fig. 1(b) [9]. Power battery provides average power, and SC provide instantaneous power when the operating conditions change instantaneously [10]. The bidirectional DC–DC converter on the power battery side realizes the voltage matching between the battery and DC bus, and the inflow of the feedback energy [11]. The SC side bidirectional DC–DC converter realizes instantaneous energy bidirectional flow and voltage matching. HESS for FC-PHEV is shown in Fig. 1(c) [12]. The power battery is used as the main energy source and the FC is used as the auxiliary energy source [13]. When the power demand of vehicle powertrain is low, only the power battery provides energy; when the power demand frequent changes, the SC provides instantaneous power; when the power battery energy is insufficient, FC starts to work. The SC provides the required instantaneous power for the powertrain and stores feedback energy.

In conclusion, considering the dynamic performances and durability of FC and power battery, most NEV adopt HESS. Firstly, the FC or power battery provides the average power required by the powertrain, the SC provides the peak power or the power required for dynamic transition, and the DC–DC converter realizes the power distribution of multiple energy sources [14]. Secondly, the output voltage of the SC in the HESS is inconsistent with the bus voltage of the vehicle, so the bidirectional DC–DC converter is required to realize the voltage matching between SC and vehicle bus. In addition, to realize energy recovery, the bidirectional DC–DC converter is required between the power battery or SC and vehicle bus to realize the flow of feedback energy. Therefore, the bidirectional DC–DC converter is the key component of HESS. It determines the performance of HESS and further affects the performance of the powertrain of NEV. Whether NEV can replace fuel vehicles and help the use of green energy and carbon reduction is closely related to the performance of bidirectional DC–DC converter in HESS of NEV.

Considering the larger variety of vehicle driving conditions, the bidirectional DC–DC converter is required to have a wide voltage conversion range to ensure that the voltage of power battery or SC and vehicle bus voltage can be effectively matched when the working conditions change. Considering the limited vehicle space, the bidirectional DC–DC converter is required to have high power density to save installation space. Considering the safety of vehicles, it is necessary to minimize the electromagnetic interference (EMI) of the powertrain, and the input and output of the bidirectional DC–DC converter should be common ground to avoid extra EMI caused by high-frequency pulse width modulation (PWM) voltage pulse [15]. Considering the damage of larger current ripple to battery and SC, the power source side current of bidirectional DC–DC converter should be continuous to improve the durability of power battery or SC [16]. Considering that the high voltage and current stresses will reduce the reliability of the converter system, the bidirectional DC–DC converter should reduce the stresses of the devices as much as possible to avoid the possibility of device damage caused by high stresses [17]. In addition, the bidirectional DC–DC converter should also have high conversion efficiency to avoid the heating and energy dissipation caused by device loss [18].

Bidirectional DC–DC converters have high requirements for power density and conversion efficiency, which need to be improved in circuit topology design and control algorithm optimization. The research on bidirectional DC–DC converter topologies plays an important role in promoting the rapid development of NEV industry. This paper systematically summarizes the bidirectional DC–DC topologies for HESS, focusing on the new topologies and novel ideas proposed in recent references, aiming to promote the research of bidirectional DC–DC converters, so as to help the use of green energy and carbon reduction.

This paper is organized as follows: Section 1 introduces the research status of existing bidirectional DC–DC topologies for HESS. In Section 2, the evaluation system of bidirectional DC–DC topologies for HESS is constructed, the comparison and evaluation of typical bidirectional DC–DC converters are performed, and suggestions for selecting the topology for HESS are provided. Section 3 summarizes the key issues of

bidirectional DC–DC converters and prospects for future research directions. Section 4 is conclusion.

2. Research status of bidirectional DC–DC topologies for HESS

The non-isolated bidirectional DC–DC converters can be obtained by replacing the power diodes in the unidirectional DC–DC converter with active switches (such as MOSFET, IGBT).

The existing bidirectional DC–DC converters mainly have the following two types: 1) Non-isolated bidirectional DC–DC converter; 2) Isolated bidirectional DC–DC converter. Non-isolated bidirectional DC–DC converter is an impedance network composed of inductor, capacitor and switch to realize direct DC–DC conversion. A transformer is added between DC and DC in the non-isolated bidirectional DC–DC converter to realize DC–AC–DC voltage conversion. A high voltage conversion ratio can be achieved by adjusting the turns ratio of the transformer. The classification of bidirectional DC–DC converters is shown in Fig. 2.

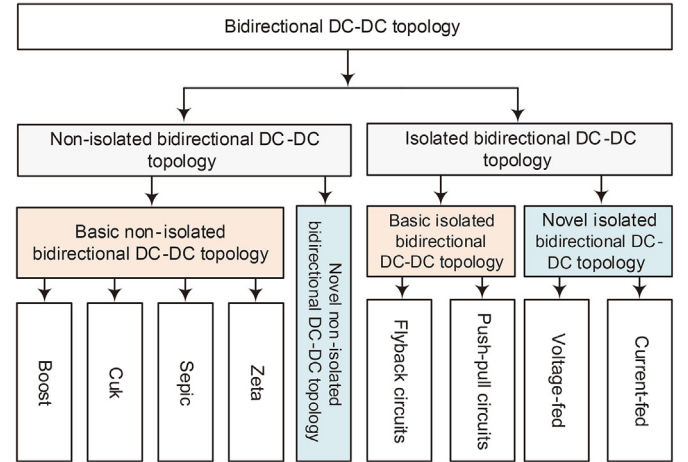


Fig. 2. Classification of bidirectional DC–DC topologies.

2.1. Non-isolated bidirectional DC–DC topologies for HESS

2.1.1. Construction of impedance networks of novel non-isolated bidirectional DC–DC converters

Based on the basic bidirectional DC–DC converters, the novel non-isolated bidirectional DC–DC converters are designed to improve the performance of bidirectional DC–DC converters through impedance transformation from the aspects of improving the voltage conversion ratio and efficiency, reducing the current ripple on the power source side and the voltage/current stresses of the devices, and improving the input and output characteristics. According to different combinations of energy storage elements, this paper creatively classifies the existing novel non-isolated DC–DC converter topologies. The novel non-isolated bidirectional DC–DC impedance networks can be divided into four types: inductor + inductor, capacitor + capacitor, inductor + capacitor and hybrid/cascaded. The classification method is shown in Fig. 3.

2.1.1.1. Inductor + inductor type. To improve the voltage conversion ratio, two inductors are charged in parallel and discharged in series, the switched inductor impedance network can be obtained. To reduce the current ripple of power source side, each inductor turns on in turn with the same trigger phase difference, and an interleaved impedance network is obtained. To improve the voltage conversion ratio, the power source and one capacitor are used as energy sources respectively to realize inductors parallel charging and series discharging, the quadratic Boost impedance network can be obtained. The derivation method of inductor + inductor impedance networks is shown in Fig. 4.

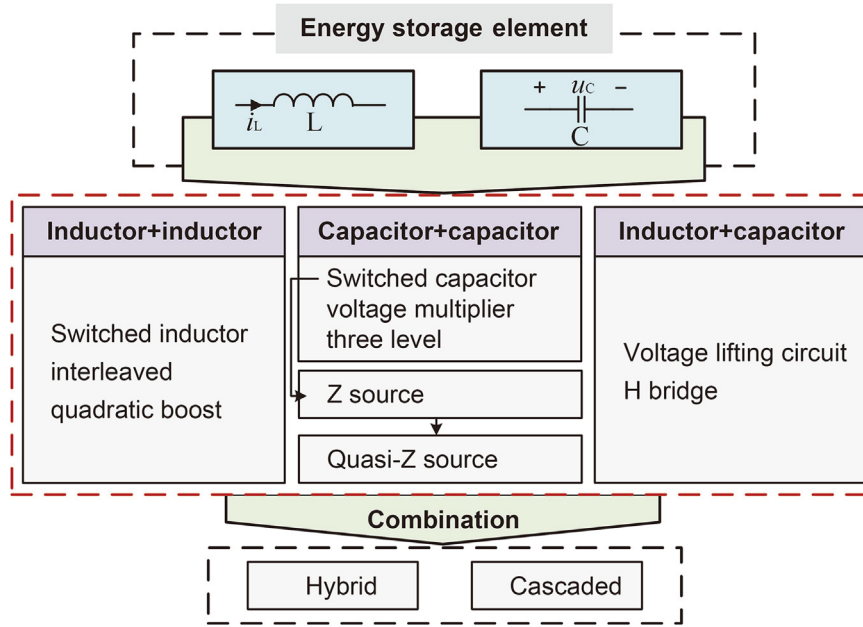


Fig. 3. Classification of novel non-isolated bidirectional DC-DC impedance networks.

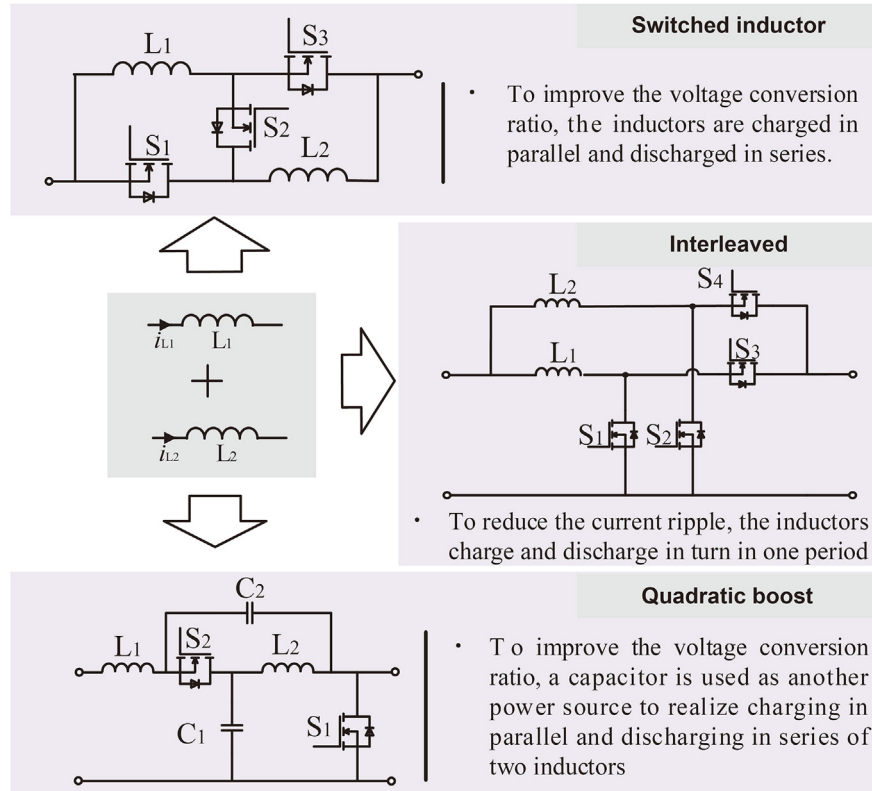


Fig. 4. The derivation of inductor + inductor impedance network.

2.1.1.2. Capacitor + capacitor type. To improve the voltage conversion ratio, two capacitors are charged in parallel and discharged in series, the switched capacitor impedance network can be obtained. To improve the voltage conversion ratio, the energy is transferred between the two capacitors firstly, and then the capacitor and the power source supply energy to the load, simultaneously, the voltage multiplier (VM) impedance network can be obtained. When the converter works stably, inductor and switch only flow current in one direction. Without violating the circuit law, the Z source impedance network can be

obtained by replacing the switch in the switched capacitor with the inductor. To overcome the defects of discontinuous current at the power source side and non-common ground of input and output of Z source, the quasi-Z source impedance network can be obtained by impedance transformation of Z source. To reduce the voltage stress, two capacitors are designed to withstand the voltage of high voltage side together, and the three level impedance network can be obtained. The derivation method of capacitor + capacitor impedance networks is shown in Fig. 5.

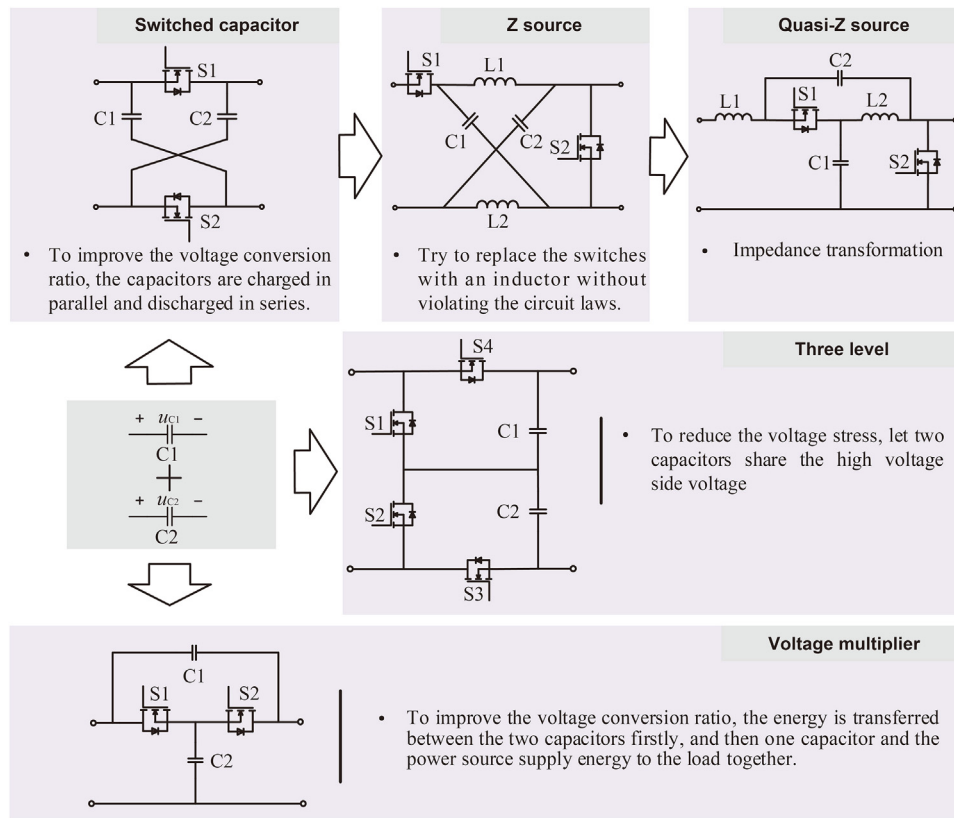


Fig. 5. The derivation of capacitor + capacitor impedance network.

2.1.1.3. Inductor + capacitor type. To improve the voltage conversion ratio, a capacitor and an inductor are charged in parallel and discharged in series to obtain the voltage lifting circuit. To improve the voltage conversion ratio, capacitors (or inductors) are connected between the

two bridge arms composed of four switches, and inductors (or capacitors) are installed on the power source side to realize filtering, forming an H-bridge impedance network. The derivation method of inductor + capacitor impedance networks is shown in Fig. 6.

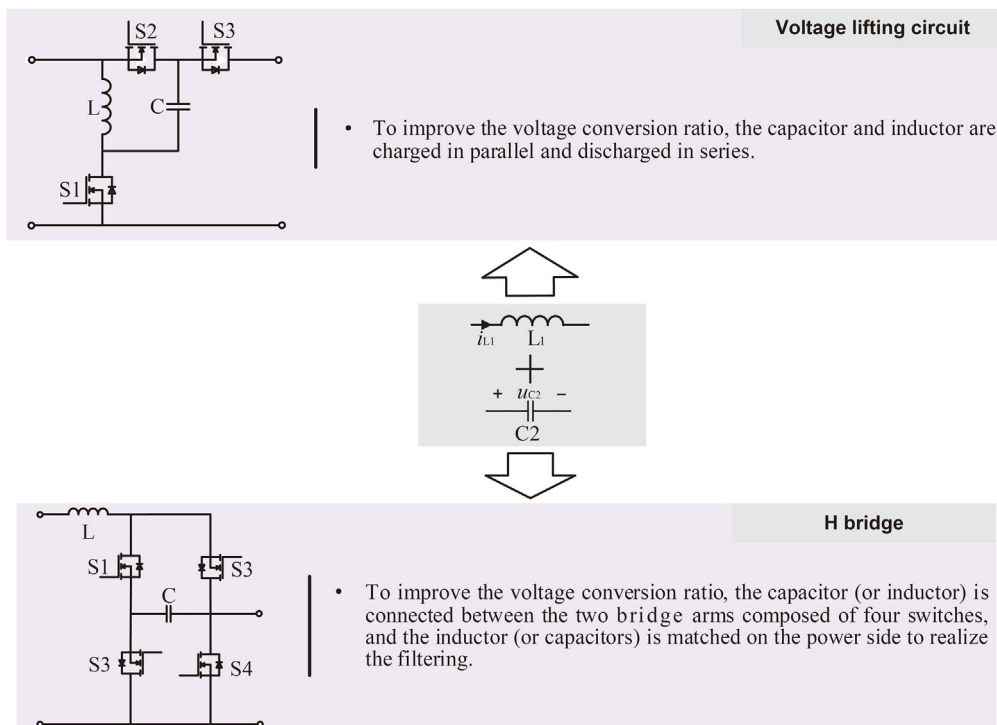


Fig. 6. The derivation of inductor + capacitor impedance network.

2.1.1.4. Hybrid/cascaded. Integrating and cascading the three types of impedance networks above can optimize the performance of bidirectional DC–DC converters and aid in obtaining novel topologies. Cascaded impedance network is to optimize the topological performance, connect more than two impedance networks in sequence, and share some devices without violating the circuit law. A hybrid impedance network can be obtained by replacing functional units of the existing topology with topologies (or impedance networks) to realize topology optimization. In the existing bidirectional DC–DC references, there are many reports on cascaded structure, and hybrid structure is relatively less. This paper focuses on the cascaded topologies.

2.1.2. Research status of existing non-isolated bidirectional DC–DC converter topologies

The derivation processes of the impedance networks of the novel bidirectional DC–DC converters are similar to that of the unidirectional DC–DC converters. The corresponding bidirectional DC–DC converter topology can be obtained by replacing the diode in the unidirectional DC–DC converter with the switch. Considering that NEV have high requirements for the input and output common ground, and power source side current ripple of bidirectional DC–DC converters, there are few topologies based on switched inductor, switched capacitor, Z source and voltage lifting circuit. In addition, the references of bidirectional DC–DC converters based on quadratic Boost and hybrid is also relatively rare. Existing non-isolated bidirectional DC–DC converter topologies include bidirectional Boost, Cuk, SEPIC topologies, interleaved bidirectional DC–DC topologies, three level bidirectional DC–DC topologies, bidirectional DC–DC topologies based on VM, bidirectional DC–DC topologies based on quasi-Z source, bidirectional DC–DC topologies based on H bridge, cascaded bidirectional DC–DC topologies. Therefore, this paper will focus on the research status of these seven kinds of topologies.

2.1.2.1. Basic bidirectional DC–DC topologies. The basic bidirectional DC–DC topologies include bidirectional Boost, Cuk and SEPIC. The derivation of bidirectional Boost impedance network is shown in Fig. 7.

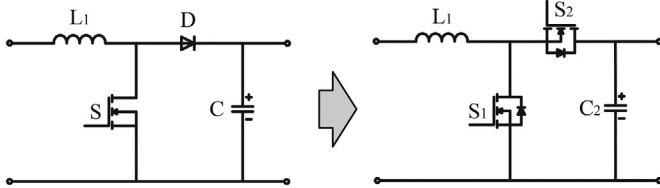


Fig. 7. The derivation of bidirectional Boost impedance network.

For the powertrain of NEV, Ref. [19] used bidirectional Boost to realize the bidirectional energy transfer between SC and power battery. Ref. [20] discussed the package and integration of bidirectional Boost converter suitable for vehicles. In Ref. [21], DC–DC converters such as bidirectional Boost, Cuk and SEPIC were analyzed and compared, so as to select the power electronic interface suitable for the vehicle powertrain. Guided by the technical specifications of bidirectional DC–DC converter, Ref. [22] comprehensively compared bidirectional Boost, Cuk and SEPIC.

The basic bidirectional DC–DC topologies have simple structure, low number of devices and cost. However, when the basic converter operates under an extreme duty cycle, parasitic parameters will significantly reduce the conversion efficiency, thus limiting the voltage conversion range. In vehicles with low SC output voltage, the basic bidirectional DC–DC converters are difficult to effectively match the SC voltage and vehicle bus voltage due to the low voltage conversion ratio.

2.1.2.2. Interleaved bidirectional DC–DC topologies. The switches of interleaved bidirectional DC–DC topologies are turned on in turn with the

same trigger phase difference. The energy was evenly distributed to the branches where the switches are located, effectively eliminating the current ripple in power source side. In addition, if the capacitor can be used to collect the energy of one branch and release energy with the other branch simultaneously, the voltage conversion ratio can be multiplied [23]. The interleaved unidirectional DC–DC converter proposed in Ref. [24] was constructed with above idea, and its voltage conversion ratio was $2/(1-d)$, which was twice that of traditional interleaved converters. The method used in Ref. [24] can be directly applied to bidirectional DC–DC converter. The bidirectional interleaved structure is suitable for the front-stage of the bidirectional topologies of HESS. The structure of interleaved bidirectional DC–DC impedance network is shown in Fig. 8.

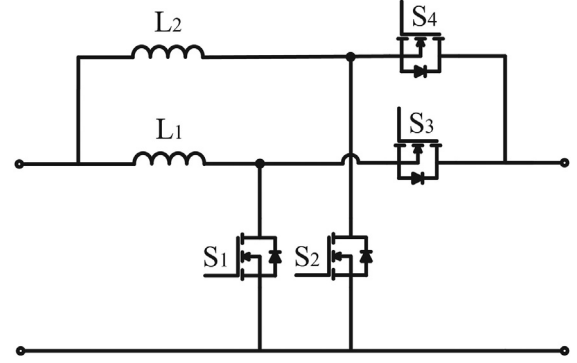


Fig. 8. Interleaved bidirectional DC–DC impedance network.

Ref. [25] discussed and tested the three phase interleaved bidirectional DC–DC converter, and proposed two kinds of bidirectional converters. Ref. [26] studied the multi-phase interleaved bidirectional DC–DC converter, and made a 30 kW converter prototype to verify feasibility. In Ref. [27], two multi-phase 1 kW DC–DC converters made of many interleaved buck phases (16 and 36) were presented. Owing to the use of surface mount technology, the prototype maximum efficiency was 95%. Ref. [28] carried out soft switching optimization for three phase interleaved bidirectional DC–DC converter, which reduced the switching loss and achieved the high efficiency of 98%. Ref. [29] proposed a novel interleaved bidirectional DC–DC converter with wide voltage conversion range, which not only had low voltage and current stresses of the switches, but also realized voltage and current self-balancing. Ref. [30] proposed an interleaved switched capacitor bidirectional DC–DC converter with high voltage conversion ratio. The converter adopted interleaved technology to reduce the current ripple of power source side, and utilized switched capacitor on the DC bus side of the converter to obtain wide voltage conversion range.

Interleaved bidirectional DC–DC topologies can significantly reduce the current ripple on the power source side of the converters, and improve the durability of the power battery or SC. However, their disadvantages are also obvious. The number of switches of interleaved topologies is large, resulting in larger volume. In addition, multiple switches turn on in turn, which also puts forward high requirements for control accuracy.

2.1.2.3. Three level bidirectional DC–DC topologies. The three level bidirectional DC–DC topology can effectively reduce the voltage stress on the devices of high-voltage side, as shown in Fig. 9. The bidirectional three level impedance network is suitable as the bus side structure of the bidirectional DC–DC converter for HESS.

The three level bidirectional DC–DC topology proposed in Ref. [31] had a wide voltage conversion range and low voltage stress of devices. The maximum efficiency of step-up and step-down was 94.1% and 94.8%, respectively. A three level DC–DC converter as a candidate for SC application was proposed in Ref. [32]. On the basis of theoretical

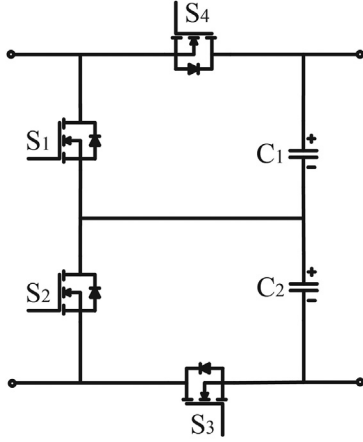


Fig. 9. Three level bidirectional DC-DC impedance network.

analysis, a 5.5 kW prototype was constructed to verify the correctness of the analysis. Ref. [33] comprehensively compared the three level bidirectional DC-DC topology with the traditional two level topology, and proved that the three level topology had higher efficiency and smaller inductor volume. Ref. [34] optimized the parallel three level bidirectional DC-DC topology, and the difference between split inductor and integrated inductor used in converter is compared in terms of weight, loss and volume. A bidirectional three level DC-DC converter and corresponding control strategies for HESS of NEV were proposed in Ref. [35], which was synthesized from Buck and Boost three level DC-DC topologies with a high voltage conversion ratio and non-extreme duty cycle.

The three level bidirectional DC-DC topologies can reduce the voltage stress of the switches to half of the high-voltage side voltage, which can improve the reliability of converter when used in the HESS. However, the three level bidirectional DC-DC topology can not significantly broaden the voltage conversion range, has a large number of switches, and the input and output are generally non-common ground. Moreover, the issue of capacitor voltage balancing also needs to be considered, which increases the design complexity.

2.1.2.4. Bidirectional DC-DC topologies based on VM. The bidirectional DC-DC topology based on VM uses two capacitors to transmit energy and can multiply the low-voltage side voltage, as shown in Fig. 10. Thus, bidirectional VM impedance network is suitable as high-voltage side structure of bidirectional DC-DC converter for HESS.

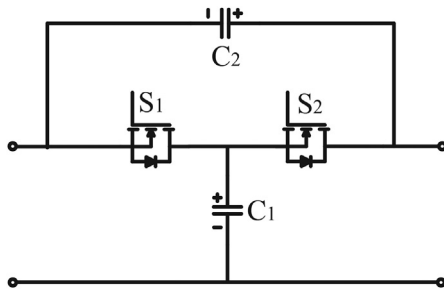


Fig. 10. Bidirectional DC-DC impedance network based on VM.

Ref. [36] proposed a bidirectional DC-DC converter based on VM with wide voltage conversion range and common ground structure. The prototype maximum efficiency was 94.45% and 94.39%, respectively. A converter proposed in Ref. [16] adopted a modified VM as the high-voltage side structure, which had a wide voltage conversion range, low current ripple. The prototype maximum efficiency was 94.2% when the rated power is 400 W. Ref. [37] adopted the bidirectional VM as the

high-voltage side structure of the proposed topology, which had the merits of wide voltage conversion range and low voltage stress. Owing to the low number of devices, the prototype maximum efficiency was 97.5%. The bidirectional DC-DC converter with high voltage conversion ratio and low voltage stress of switches was obtained in Ref. [38] by cascading the switched inductor with VM.

Bidirectional VM has the merits of wide voltage conversion range and simple structure. It is suitable as the high-voltage side structure of the converter for HESS. However, the voltage stress on the high voltage side is high, which increases the possibility of device defects and leads to the low reliability of the powertrain. Therefore, when the bidirectional topologies based on VM is applied to HESS, the scheme should be optimized from the perspective of reducing voltage stress of the devices.

2.1.2.5. Bidirectional DC-DC topologies based on Quasi-Z source. The bidirectional quasi-Z source impedance network is derived from the impedance transformation of the Z source. The bidirectional quasi-Z source impedance network modifies the issues of discontinuous current at the power source side and non-common ground of the Z source. It is suitable as the low-voltage side structure of the bidirectional DC-DC topology for HESS. The bidirectional quasi-Z source impedance network is shown in Fig. 11.

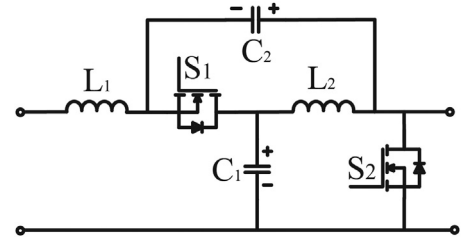


Fig. 11. Bidirectional DC-DC impedance network based on quasi-Z source.

Ref. [39] proposed a cascaded bidirectional quasi-Z source DC-DC converter, which had stronger voltage conversion capability and lower current ripple. Ref. [9] changed the MOSFETs' position of the traditional two level quasi-Z source bidirectional DC-DC converter, and the obtained new topology had wide voltage conversion range, low voltage stress and common ground characteristics. In Ref. [10], a bidirectional switched capacitor was added to the last stage of the topology proposed in Ref. [9] to obtain a wider voltage conversion range and lower voltage stress of switches. Ref. [40] compared the traditional bidirectional Boost converter with quasi-Z source converter in four aspects: capacitor energy, inductor energy, voltage stress and conduction loss. The results shown that quasi Z-source bidirectional DC-DC had more merits.

The bidirectional quasi-Z source converter has the same voltage conversion ratio as the Z source converter. In theory, high voltage conversion ratio can be obtained when the duty cycle is close to 0.5, which avoids the decrease of conversion efficiency caused by parasitic parameters under the extreme duty cycle. In addition, the power source side current of the bidirectional quasi-Z source impedance network is continuous and the input and output are common ground, which is suitable for HESS of NEV. However, the duty cycle range of quasi-Z source impedance network within 0–0.5. When the duty cycle is close to 0.5, the slope of voltage conversion ratio is steep, which puts forward high requirements for control accuracy.

2.1.2.6. Bidirectional DC-DC topologies based on H bridge. The H bridge bidirectional DC-DC impedance network use four switches to form a pair of bridge arms, and energy storage elements are arranged between the two bridge arms to realize the bidirectional flow of energy, as shown in Fig. 12. H bridge impedance network is suitable as high voltage side structure of bidirectional DC-DC converter for HESS.

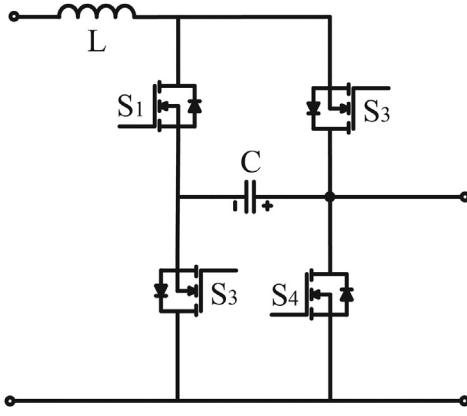


Fig. 12. Bidirectional DC-DC impedance network based on H bridge.

For the background of HESS, a novel common grounded H bridge bidirectional converter characterized by a good dynamic response, a low inductor current ripple, and a wide voltage conversion ratio was proposed in Ref. [41]. In addition, it can avoid the narrow pulse of PWM voltage waveforms when a high voltage conversion ratio was achieved. Ref. [42] presented a wide voltage conversion range bidirectional DC-DC converter with a synchronous rectification H Bridge. The high ratio bidirectional operations can be obtained by non-extreme duty cycle that were near 0.5, which avoided the efficiency decrease caused by extreme duty cycle. Ref. [43] improved the topology proposed in Ref. [42], which not only inherited the advantages of the topology in Ref [42], but also obtained the advantage of common ground. Ref. [44] designed a parallel bidirectional

DC-DC converter based on H bridge, which had simple structure and high efficiency in wide load range. The prototype maximum efficiency was 97.8%. Ref. [45] presented a novel high voltage conversion ratio bidirectional DC-DC converter. The converter did not require the operation at extremely higher duty cycle ratio and the configuration was simple. However, this converter was non-common ground structure.

The H bridge bidirectional DC-DC converter has a less number of energy storage elements and is easy to achieve high power density. A high voltage conversion ratio can be obtained when the duty cycle is close to 0.5. However, the large number of switches leads to the relatively complex control strategies and circuits. In addition, similar to the quasi-Z source structure, the slope of voltage conversion ratio is steep when duty cycle is close to 0.5, which puts forward high requirements for control accuracy.

2.1.2.7. Cascaded bidirectional DC-DC topologies. Cascaded bidirectional DC-DC topologies can be obtained by cascading two or more bidirectional DC-DC topologies as required (repeated devices can be shared), as shown in Fig. 13. Cascaded method is an important way to construct bidirectional DC-DC converters for HESS. Many novel bidirectional DC-DC topologies can be regarded as a cascade structure, as shown in Table 1.

It can be seen from Table 1 that the cascaded structure can combine sub-topologies according to the application requirements to obtain a novel topology, and inherit the advantages of each sub-topology. The disadvantages of cascaded bidirectional DC-DC converters are that the number of devices and the converter volume are large. The efficiency of cascaded converters is the product of that of each stage sub-converter, and the overall efficiency will not be high [47]. In addition, the cascaded topologies will inherit the shortcomings of each sub-topology.

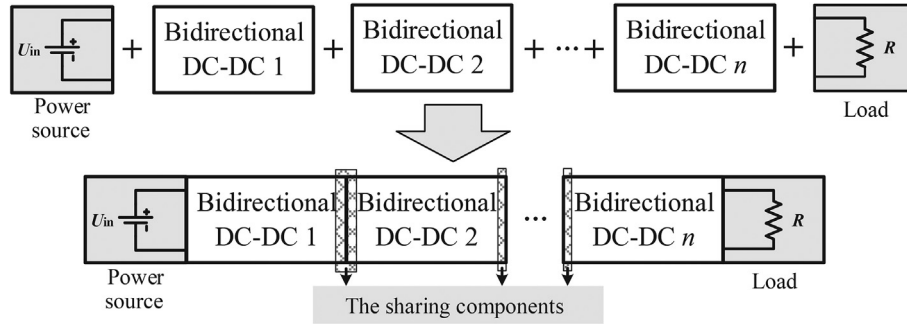


Fig. 13. Schematic diagram of bidirectional cascaded topologies.

Table 1
Composition and Advantages of Cascaded bidirectional DC-DC Topologies.

Reference	Sub-topology 1	Sub-topology 2	Advantages	Disadvantages	Applicable to HESS
Ref. [38]	Switched inductor	VM	- Wide voltage conversion range - Low voltage stress	- Input-output non-common ground - High current ripple in power side	★
Ref. [37]	Boost	VM	- Low voltage stress - Input-output common ground - Low current ripple in power source side - Low voltage stress	Low voltage conversion ratio	★★
Ref. [32]	Boost	Three level	- Low current ripple in power source side - Low voltage stress - Input-output common ground - Low current ripple in power source side	- Low voltage conversion ratio - Input-output non-common ground - Low voltage conversion ratio	★★
Ref. [39]	Quasi-Z source	Quasi-Z source	- Low current ripple in power source side - Wide voltage conversion range - Low current ripple in power source side - Low voltage stress	- Narrow duty cycle range - Input-output non-common ground - High control requirements - Narrow duty cycle range	★★★
Ref. [30]	Interleaved	Boost/VM	- Wide voltage conversion range - Low current ripple in power source side - Low voltage stress - Wide voltage conversion range - Low current ripple in power source side - Low voltage stress	Input-output non-common ground	★
Ref. [10]	Quasi-Z source	Switched capacitor	- Wide voltage conversion range - Low current ripple in power source side - Low voltage stress - Wide voltage conversion range - Low current ripple in power source side - Low voltage stress	- Input-output in reverse polarity - High current ripple in power source side	★
Ref. [46]	Switched inductor	VM	- Wide voltage conversion range - Low voltage stress - Input-output common ground		

Note: ★ indicates particularly inappropriate, ★★ indicates relatively appropriate, ★★★ indicates particularly appropriate.

Table 2 systematically summarizes the advantages and disadvantages of typical non-isolated bidirectional DC–DC impedance networks, which provides a reference for the construction of bidirectional DC–DC converters for HESS of NEV.

2.2. Isolated bidirectional DC–DC topologies for HESS

Compared with the non-isolated bidirectional DC–DC converter, the high-frequency transformer can ensure the galvanic isolation of the input and output in the bidirectional topology for HESS, which can adjust the voltage conversion range by changing the turns ratio. Therefore, the isolated bidirectional DC–DC topologies have higher voltage conversion ratio, and can match different voltage levels for HESS. The isolated bidirectional DC–DC converters can be divided into voltage-fed and current-fed according to different DC power sources, as shown in Fig. 14. Typical voltage-fed bidirectional DC–DC converter include basic converters, dual active bridge (DAB) converters, Z source and quasi-Z source full-bridge converters, interleaved isolated converters. Typical current-fed bidirectional DC–DC converter include basic current-fed converters, current-fed DAB converters and interleaved isolated current-fed converters.

Voltage-fed bidirectional DC–DC converter consists of rectifier and inverter units. The structure of the current-fed and voltage-fed converters are similar. The main differences are that the DC source types and

current-fed converter needs to add an inductor on the DC current source side to suppress current ripple. Considering that the current-fed converter is less used in vehicle applications and the current source can be replaced by a voltage source and an inductor, this paper mainly discusses the voltage-fed converter.

2.2.1. Basic voltage-fed isolated bidirectional DC–DC converter

Typical basic voltage-fed bidirectional DC–DC converters are the bidirectional flyback and push–pull converter, as shown in Fig. 15.

Ref. [48] proposed a bidirectional flyback converter, which used a high-frequency transformer with unidirectional excitation. Owing to the utilization rate of the transformer is low, it was suitable for low power level application. Ref. [49] comprehensively analyzed the operation principle, conversion capability and soft-switching realization of 12 different operation states of the push–pull converter.

Bidirectional flyback converters are widely used in low-power applications requiring galvanic isolation, due to their simple structures, low number of devices, and fast dynamic response. Since the transformer in the flyback converter needs to achieve galvanic isolation and store energy simultaneously, the switches need to withstand high voltage and current stresses. The switching losses are large, and the conversion efficiency is low. Owing to the high-frequency transformer of push–pull converters has leakage inductance, a large voltage spike will be

Table 2

Advantages and disadvantages of typical non-isolated bidirectional DC–DC impedance networks.

Classification	Impedance network	Advantages	Disadvantages	Applicable to HESS
Basic	Boost/SEPIC/Cuk	- Simple structure - Easy to control - Easy to generate new topologies - Low number of devices	Narrow voltage conversion range	★★ Suitable for low voltage side of DC–DC for HESS
Inductor + inductor	Switched inductor	- Wide voltage conversion range - Easy to control - Easy to generate new topologies	Pulsating current of power source side	★
	Interleaved	- Low current ripple of power source side - Input-output common ground - Wide voltage conversion range - Continuous power source side current - Easy to control - Input-output common ground - Wide voltage conversion range	- Requiring high control precision - Large number of switches	★★★ Suitable for low voltage side of DC–DC for HESS
	Quadratic boost	- Wide voltage conversion range - Continuous power source side current - Easy to control - Input-output common ground - Wide voltage conversion range	High voltage stress at high voltage side	★★★ Suitable for low voltage side of DC–DC for HESS
Capacitor + capacitor	Switched capacitor	- Input-output common ground - Wide voltage conversion range	- Input-output non-common ground - Pulsating current of power source side - High voltage stress at high voltage side	★
	VM	- Wide voltage conversion range - Input-output common ground	High voltage stress at high voltage side	★★★ Suitable for high voltage side of DC–DC for HESS
	Z Source	High voltage conversion ratio in non-extreme duty cycle	- Discontinuous power source side current - Input-output non-common ground - Steep slope of voltage conversion ratio - Requiring high control precision	★
	Quasi-Z source	- High voltage conversion ratio in non-extreme duty cycle - Continuous power source side current - Input-output common ground - Low voltage stress at high voltage side	- Requiring high control precision - Steep slope of voltage conversion ratio - High voltage stress at high voltage side	★★ Suitable for high voltage side of DC–DC for HESS
	Three level	- Input-output common ground - Low voltage stress at high voltage side	- Requiring high control precision - Large number of switches - No obvious increase in voltage conversion ratio - Input-output non-common ground - No obvious increase in voltage conversion ratio - Pulsating current of power source side	★★ Suitable for high voltage side of DC–DC for HESS
Inductor + capacitor	Voltage lifting circuit	- Easy to generate new topologies - Input-output common ground	- Input-output non-common ground - No obvious increase in voltage conversion ratio - Pulsating current of power source side	★
	H bridge	- Easy to generate new topologies - Less number of energy storage devices - Input-output common ground - Wide voltage conversion range - Easy to generate new topologies	- Large number of switches - Requiring high control precision	★★ Suitable for high voltage side of DC–DC for HESS
Hybrid/Cascaded	Cascaded	- Input-output common ground - Wide voltage conversion range - Easy to generate new topologies	- Large number of devices - Low conversion efficiency - Low power density	★★★
	Hybrid	- Wide voltage conversion range - Easy to generate new topologies	- Large number of devices - Low conversion efficiency - Low power density	★★★

Note: ★ indicates particularly inappropriate, ★★ indicates relatively appropriate, ★★★ indicates particularly appropriate.

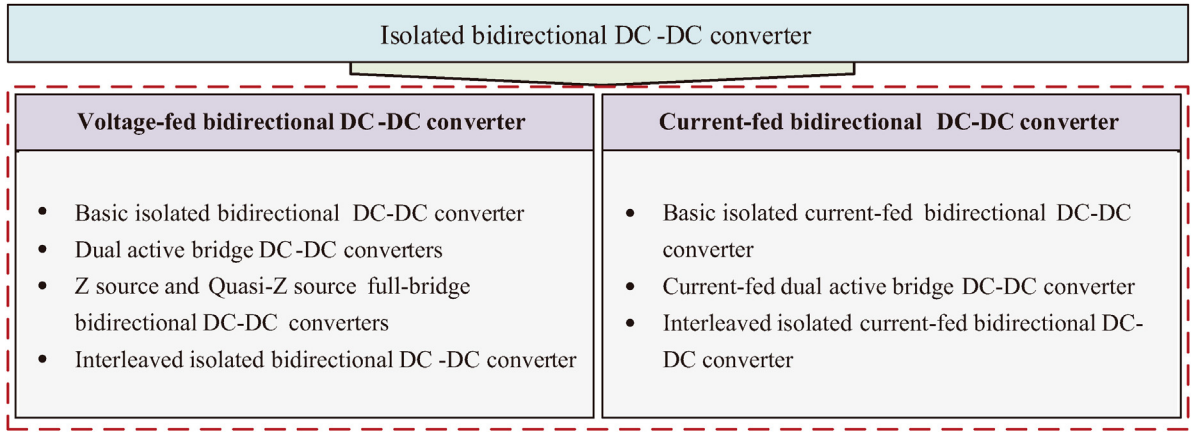


Fig. 14. Classification of isolated bidirectional DC-DC converters.

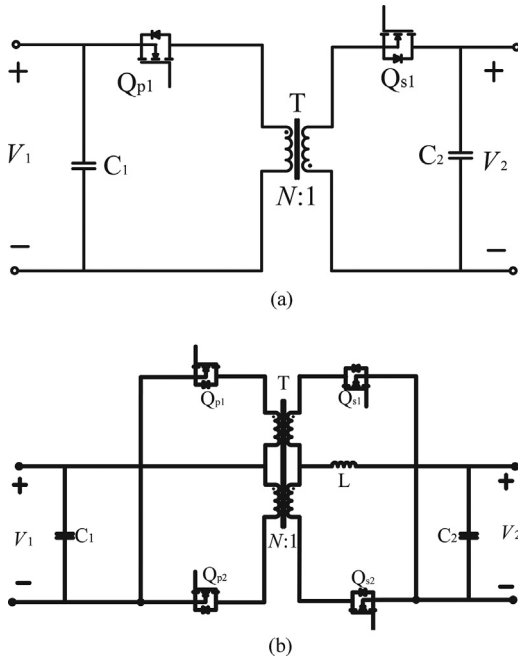


Fig. 15. Basic bidirectional DC-DC converters. (a) Bidirectional flyback converter. (b) Bidirectional push-pull converter.

generated. Moreover, the transformer has the center tap, which increases the design difficulty and reduces the reliability of the converter. The basic voltage-fed isolated converters are suitable for low-power applications, excluding HESS of NEV.

2.2.2. DAB bidirectional DC-DC topologies

According to the different configurations of primary and secondary sides of the high-frequency transformer, DAB converters can be classified as resonant and non-resonant types. For resonant type DAB, the high-frequency transformer is connected in series with a resonant circuit, which has good soft switching characteristics. The structure of resonant type DAB converters is shown in Fig. 16.

Resonant DC-DC converters have received widespread attention, owing to their advantages in reducing switching loss. Ref. [50] proposed a LC resonator structure, as shown in Fig. 17(a), the resonator was formed by an inductor and a capacitor in series, having the effect of suppressing the DC bias. Adding the magnetizing inductor to the LC resonant circuit, Ref. [51] constituted the LLC structure, which had two control methods: frequency modulation and PWM, as shown in Fig. 17(b). The LLC structure can suppress current harmonics, further

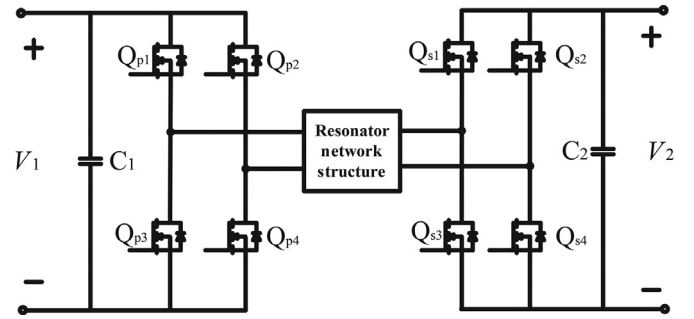


Fig. 16. Resonant DAB converter.

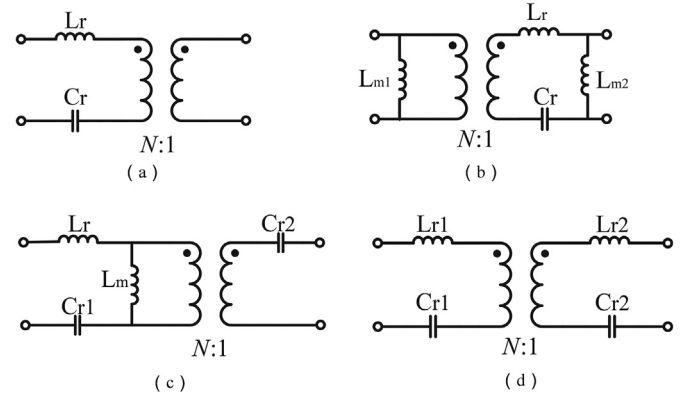


Fig. 17. Structures of the resonator. (a) LC resonator. (b) LLC resonator. (c) CLLC asymmetrical resonator. (d) CLLC symmetrical resonator.

enhance the operating frequency, and reduce the turn-off current of the switches. The CLLC symmetric and asymmetric structures were discussed in Refs. [52,53], respectively, as shown in Fig. 17(c) and (d).

For non-resonant DAB converters, the high-frequency transformer only has one inductor in series. Generally speaking, DAB converters refers to non-resonant DAB converters. The structure is divided into half bridge and full bridge types, as shown in Fig. 18.

The half bridge structures are widely used in battery charger and UPS, owing to the advantages of less number of devices and small volume. A comprehensive steady-state analysis of the dual active half-bridge converter was presented in Ref. [54]. Ref. [55] studied the typical voltage-fed dual active full-bridge converter. The number of switches of the dual active full-bridge converter is twice that of half-bridge converter [56].

The intermediate stage of the non-resonant DAB converter is only an inductor, which can not only realize the bidirectional transmission of

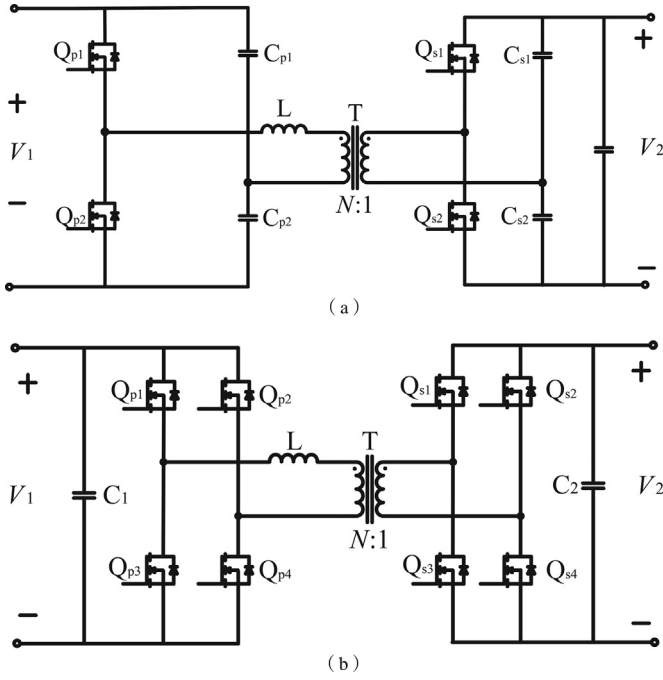


Fig. 18. Non-resonant DAB converters. (a) Dual active half-bridge DC-DC converter. (b) Dual active full-bridge DC-DC converter.

energy, but has the advantages of low voltage stress and flexible control. The resonant DAB DC-DC converter, inheriting the advantages of the DAB converter, can improve the soft switching capability and efficiency. Moreover, the resonant structure can be selected according to different application requirements. However, with the increase of resonant elements, the complexity of the converter and the difficulty of controller design increase accordingly. The bidirectional full-bridge converters are suitable for high voltage and power occasions, including HESS of NEV.

2.2.3. Z source and Quasi-Z source full-bridge bidirectional DC-DC converters

The Z source and quasi-Z source impedance network are composed of two capacitors and two inductors. The Z source impedance network is shown in Fig. 19.

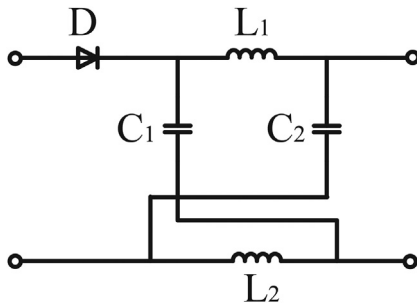


Fig. 19. Z-source impedance network.

The Z source full-bridge DC-DC converter, as shown in Fig. 20, is to add a Z source impedance network between the power source and the inverter bridge arms of the traditional full-bridge DC-DC converter. On this basis, thyristors are installed between the power source and the Z source structure to restrict the unidirectional flow of current [57]. In Ref. [58], the Z source full-bridge structure was investigated, which used the same switches to step-up, and turned the energy

shoot-through state into a operating mode of the converter. Therefore, shoot-through state caused by EMI would not damage to the converter. To make the Z source structure meet the requirements of bidirectional energy transmission, four thyristors D_{11} , D_{13} , D_{21} , and D_{23} are introduced. The thyristor D_{11} is turned off in the forward power flow and turned on in the reverse power flow to provide reverse current. The thyristor D_{13} is turned on in the forward power flow and turned off in the reverse power flow to divide the Z source network into LC filters. The functions of thyristors D_{21} and D_{23} are similar to those of thyristors D_{11} and D_{13} . Since the states of D_{11} , D_{13} , D_{21} and D_{23} change only when the current flow direction changes, the switching frequency is low, and the use of thyristors can reduce the cost and increase the reliability.

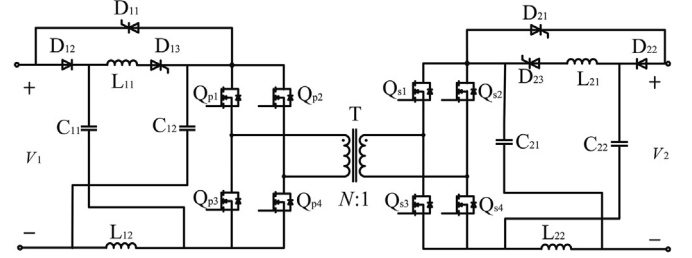


Fig. 20. Z source bidirectional full-bridge DC-DC converter.

Considering the discontinuous current at the power source side of Z-source full-bridge DC-DC converters. Ref. [59] proposed a quasi-Z source full-bridge structure, as shown in Fig. 21. In Fig. 21, the function of thyristors D_{11} , D_{13} , D_{21} , D_{23} is similar to that of Z source bidirectional full-bridge DC-DC converter.

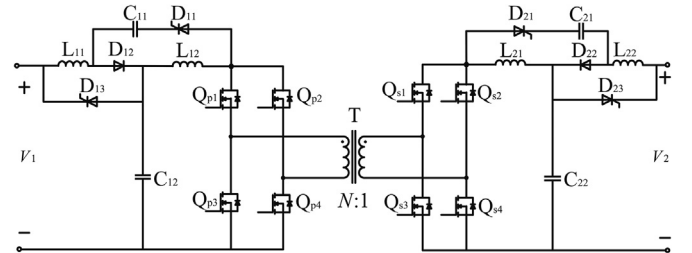


Fig. 21. Quasi-Z source bidirectional full-bridge DC-DC converter.

The Z source network is added to the typical DAB structure, which can effectively eliminate the danger of the shoot-through of the same bridge arm and improve the conversion efficiency. Since the startup current will reduce the service life of devices and system stability. The quasi-Z source full bridge structure can improve the startup current and the issue of current discontinuity of Z-source full bridge structure. However, the above modified converters increase the number of devices and the design difficulty, and reduce system reliability, so it is not suitable for HESS of NEV.

2.2.4. Interleaved isolated bidirectional DC-DC converter

The interleaved structure can not only share the voltage and current stresses, reduce the current ripple, but raise the voltage conversion ratio and power transmission level. The interleaved isolated converters are divided into Y-Y and $\Delta - \Delta$ types, and the resonant structure of interleaved converters is similar to the DAB resonant structure.

An interleaved current-fed bidirectional full-bridge DC-DC converter was proposed in Ref. [60], as shown in Fig. 22, which had the advantages of low input voltage stress and low output current stress, as well as reducing the transformer turn ratio and output side current ripple.

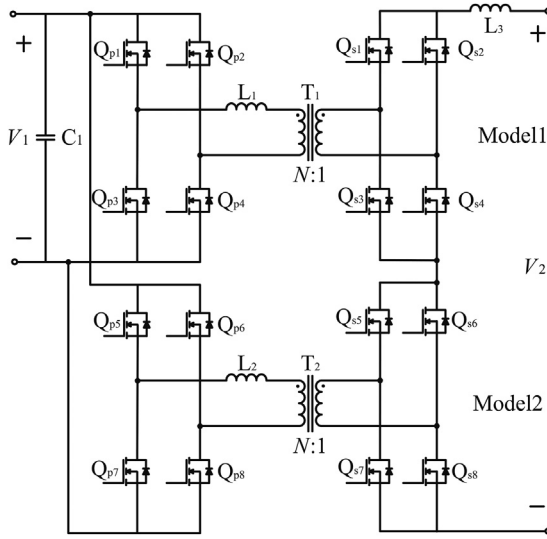


Fig. 22. Two phase interleaved bidirectional DC-DC converter.

Ref. [61] studied the operating characteristics of the $\Delta-\Delta$ three phase interleaved LLC resonant DC-DC converter. The inverter part of the topology was composed of two three level bridge arms, as shown in Fig. 23. The $\Delta-\Delta$ structure was used to connect the three phases, which can achieve good power distribution, easy thermal management, and low input and output current ripples, helping to achieve high-power applications and facilitate device selection. Ref. [62] studied the operating characteristics of the three phase Y-Y type interleaved LLC resonant DC-DC converter, as shown in Fig. 24. The Y-Y type connection structure could reduce the power source side current ripple, and the current self-balancing could be realized [63].

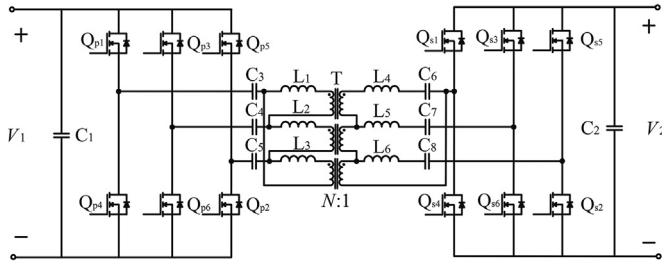


Fig. 23. Three phase $\Delta-\Delta$ interleaved LLC resonant bidirectional DC-DC converter.

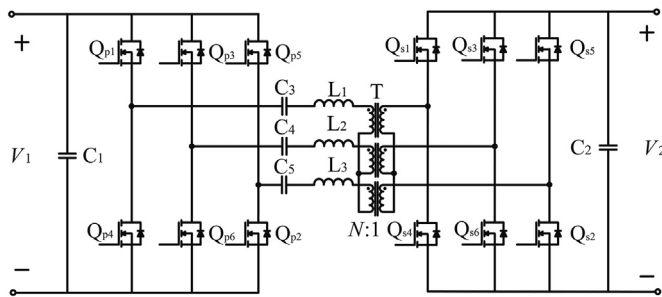


Fig. 24. Three phase Y-Y interleaved LLC resonant bidirectional DC-DC converter.

The two phase dual active full-bridge structure has the advantages of low input and output current ripples. Moreover, they can reduce the current stress on the primary side devices, the voltage stress on the secondary side devices, and the turn ratio of the transformer. However,

there are the issues of a large number of devices and current balancing requirement, which increases the design difficulty. The three phase interleaved LLC resonant full-bridge structure can adjust the voltage conversion ratio through phase shift and frequency modulation control strategies, which is suitable for applications requiring large difference in input and output voltages. Among them, Y-Y structure is widely concerned for its outstanding current self-balancing capability.

Table 3 systematically summarizes the advantages and disadvantages of typical isolated bidirectional DC-DC converters, which provides a reference for the construction of bidirectional DC-DC converters for NEV.

Since the structures of voltage-fed and current-fed are similar, and considering that current-fed bidirectional DC-DC converter is rarely used in HESS, this paper mainly discusses the voltage-fed isolated bidirectional DC-DC converter, and does not further analyze the current-fed converter. The current-fed converter has the advantages of low current ripple and high voltage conversion ratio. However, it also has the disadvantages of large peak current under light load, the large power loss, and existing voltage spikes of switches. Compared with the current-fed, the voltage-fed bidirectional DC-DC converters have the advantages of low device stress, good stability and small volume. However, it also has the disadvantages of large current ripples and narrow voltage conversion range.

Owing to the magnetic flux leakage in the transformer will produce voltage spikes in the converter, additional absorption circuits are needed to suppress voltage spikes, which increases the cost and volume of the converter. In addition, the transformer operating at high frequency is an EMI source, which seriously affects the safe operation of the bidirectional DC-DC converter. Therefore, the transformerless bidirectional DC-DC converter is more suitable for NEV. Considering the reliability, cost, installation space and other factors of NEV, the existing HESS for NEV mostly use the non-isolated bidirectional Boost topology [11], anti-parallel structure of Buck and Boost [64], and cascaded structure of bidirectional Buck and Boost [65]. In addition, an interleaved bidirectional DC-DC converter can effectively reduce the power side current ripple and improve the service life of power battery or SC. Gao Dawei's team from Tsinghua University successfully applied an interleaved converter to vehicle bus and obtained favorable results [66]. In the scenario where galvanic isolation of input and output is required, considering the reliability and maturity of the system, bidirectional half-bridge and full-bridge topologies are used [64].

3. Evaluation of bidirectional DC-DC converters for HESS

3.1. Evaluation indexes of non-isolated bidirectional DC-DC converter for HESS

The evaluation indexes of the bidirectional DC-DC converters constitute the standard based on which the merits and demerits of the topology are assessed. The evaluation indexes of bidirectional DC-DC converter for HESS include voltage conversion ratio (step-up ratio, step-down ratio), current ripple of power source side, voltage and current stresses, conversion efficiency, power density, number of devices, slope of voltage conversion ratio, and presence of common ground. Based on the above evaluation indexes, a novel evaluation system of bidirectional DC-DC converter is constructed.

3.1.1. Voltage conversion ratio

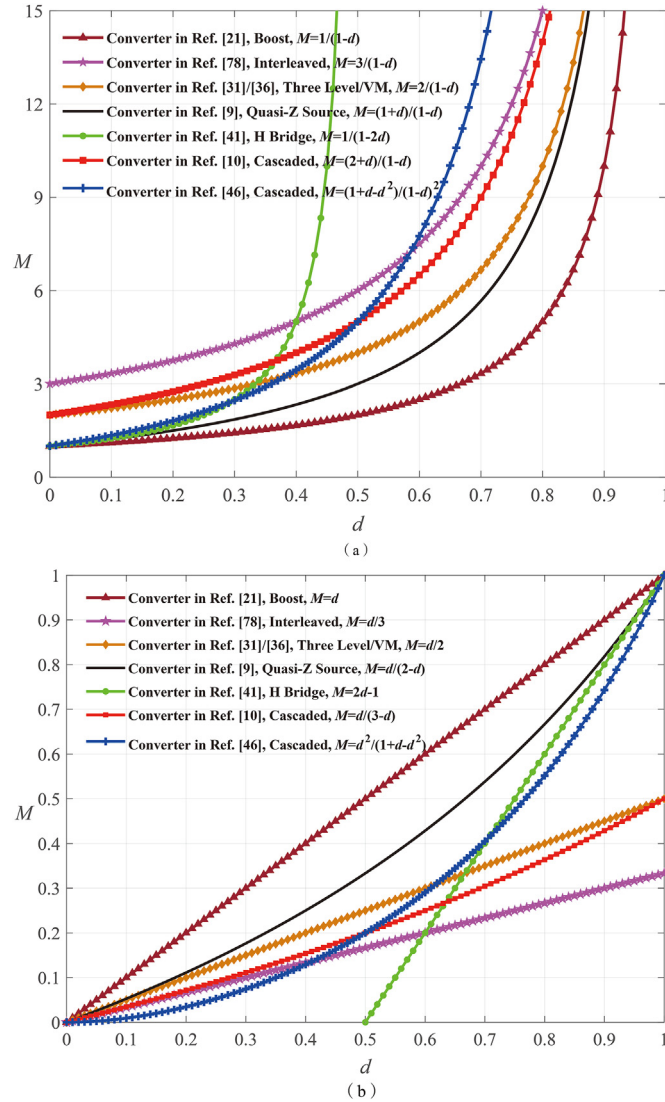
Voltage conversion ratio is the ratio of output voltage to input voltage of bidirectional DC-DC topologies. Owing to the bidirectional flow of energy, there is a step-up ratio and a step-down ratio. The voltage conversion ratio is the most important index to evaluate the bidirectional DC-DC converters. The strength of the voltage conversion capability of the topology increases with the voltage conversion ratio. Considering that the vehicle driving conditions change greatly, the bidirectional DC-DC converter is required to have a wide voltage conversion range. Fig. 25 compares the voltage conversion ratios of seven typical bidirectional DC-DC converters.

Table 3

Advantages and disadvantages of isolated bidirectional DC–DC converters.

Classification	Advantages	Disadvantages	Applicable to HESS
Basic	- Simple structure - Easy to control - Easy to generate new topologies - Low number of devices	- Large switching loss - Low conversion efficiency - Low power applications	★
DAB	- Wide voltage conversion range - Low stress of switches - Flexible control method - High power density	- Complex structure - Difficult to control - Large volume and cost	★★
Z source/Quasi-Z source	- High conversion efficiency - Small startup current	- Complex structure - Large number of devices - Difficulty to design	★
Interleaved	- Wide voltage conversion range - Low current ripple - High power density	- Large number of switches - Complex structure	★★

Note: ★ indicates particularly inappropriate, ★★ indicates relatively appropriate.

**Fig. 25.** Voltage conversion ratio curves of typical bidirectional DC–DC topologies. (a) step-up ratio curves. (b) step-down ratio curves.

3.1.2. Current ripple of power source side

During the high-frequency switching of bidirectional converters, the internal energy storage elements are also charged and discharged at high frequency. This would, in turn, generate current ripples at the power source side of the converter. Since the low-voltage side of the bidirectional

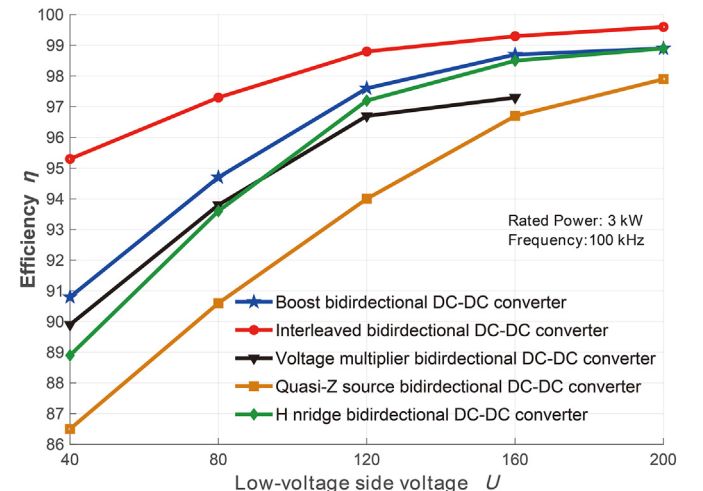
DC–DC converters is directly connected to the SC or battery, if the current ripple on the power source side of the converters is large, the service life of the battery or SC and conversion efficiency of the converters are affected [16]. Although the current ripple can be filtered using filters, the low-voltage side current of the bidirectional DC–DC converters for NEV should not be pulsating considering the large filter volume.

3.1.3. Voltage and current stress

Voltage stress is the maximum reverse voltage withstand by the device when it is turned off. Current stress is the maximum current flowing through the device. The significant increase in the device stress, in turn, increases the possibility of device defects, leading to low reliability of the converter. In addition, the voltage and current stresses determine the rated parameters of power electronic devices (e.g., drain-source voltage V_{DS} , continuous drain current I_D). Generally, the higher the voltage and current stress the device can withstand, the more expensive and the larger the volume. To reduce the cost and volume, the power electronic devices in the bidirectional DC–DC converters for HESS should have low voltage and current stresses.

3.1.4. Conversion efficiency

Conversion efficiency is the ratio of output power to input power. Owing to the parasitic parameters and the frequent switching on and off of the switches, bidirectional DC–DC converter losses include conduction and switching losses, which not only results in an output power lower than the input power but also raises the temperature of the converter. The heating caused by low efficiency not only causes the waste of energy, but requires a large volume cooler, so the bidirectional DC–DC converter

**Fig. 26.** Conversion efficiency curves of typical bidirectional DC–DC topologies.

needs to pursue high conversion efficiency. This paper compares the efficiency of typical bidirectional DC–DC converters when the input voltage is 40 V–200 V, the output voltage is 400 V, the rated power is 3 kW and the switching frequency is 100 kHz. The efficiency curves are shown in Fig. 26. When the power is constant and the input voltage declines, the efficiency decreases correspondingly, owing to the increasing losses caused by the increasing input current.

3.1.5. Power density

The ratio of the power to the volume of the bidirectional DC–DC converter is defined as the power density. Owing to the limited space in NEV, the converter is required to process high power at a small volume. High-frequency is the most effective methods of increasing the power density. However, with increasing operating frequency of the device, severe EMI issues begin to arise.

3.1.6. Number of devices

The number of devices is the sum of inductors, capacitors and switches in bidirectional DC–DC converters. A large number of devices increase the volume of the DC–DC converters and reduce the power density. Cascaded bidirectional DC–DC topologies can general improve the performance of a certain aspect, but there is a issue of large number of devices. The number of devices should be reduced as much as possible to accommodate the limited vehicle space.

3.1.7. Slope of voltage conversion ratio

The slope of the voltage conversion ratio is an index that describes the change in the voltage conversion ratio with the duty cycle. A steep voltage conversion ratio slope implies that the voltage conversion ratio increases rapidly with a slight increase in the duty cycle. The designers of the bidirectional DC–DC converter tend to favor a gentle slope of the voltage conversion ratio for it indicates that the control requirement of the converter is easy to achieve. The voltage conversion ratio slope is the slope of each curve in Fig. 25.

3.1.8. Common ground

The du/dt and di/dt caused by switches are the main EMI sources in the bidirectional DC–DC converters [67]. When the bidirectional DC–DC converter does not have a common ground structure, the high-frequency PWM voltage difference would generate extra EMI, which will affect the safe operation of the converter and vehicle. Therefore, common ground is a key index in evaluating the bidirectional DC–DC converter for HESS.

3.2. Evaluation indexes of isolated bidirectional DC–DC converter for HESS

For isolated bidirectional DC–DC converters, voltage conversion ratio, power source side current ripple, voltage and current stresses, conversion efficiency, power density, number of devices and the slope of voltage conversion ratio are also important evaluation indexes. However, owing to the existence of transformer, the winding design of transformer needs to be considered.

Transformer winding design: as the core component of isolated bidirectional DC–DC converter, the design difficulty, utilization rate and volume of transformer directly affect the cost, efficiency and power density of the converter. When evaluating the topology, it is necessary to consider the design of the transformer winding, and ensure that the design difficulty and volume are as small as possible, while the utilization rate is as high as possible.

3.3. Comparison and evaluation of typical non-isolated bidirectional DC–DC converter

In this section, non-isolated bidirectional DC–DC converters are compared in terms of ten aspects: the technology used, power source side current, the step-up ratio, the step-down ratio, duty cycle range, voltage stress of switches, current stress of switches, number of devices, and presence of common ground. The distribution of indexes for nine typical non-isolated bidirectional DC–DC topologies are summarized in Table 4.

To compare the bidirectional DC–DC converter more intuitively, and aid the topology design of HESS, this study qualitatively evaluated eight typical non-isolated bidirectional DC–DC topologies in terms of eight aspects: power side current ripple, step-up ratio, step-down ratio, voltage stress, current stress, number of devices, the slope of the voltage conversion ratio, and presence of common ground, as shown in Fig. 27. Three points are to be explained: firstly, the advantages and disadvantages of the topology are relative. The topologies evaluated in Fig. 27 only reflect the relationship between eight topologies. Secondly, no topology is excellent in all aspects. Performances optimized of the topology should be conducted from the perspective of application when designing the topology. Thirdly, the power density and conversion efficiency of converters are affected by many factors, such as the operating frequency, rated power, and the number of devices, which cannot be simply compared.

Table 4
Indexes of typical non-isolated bidirectional DC–DC topologies.

Topologies	Technology used	Power source side current	Step-up ratio U_{high}/U_{low}	Step-down ratio U_{low}/U_{high}	Duty cycle range	Voltage stress of switches	Current stress of switches	Number of devices			Common ground
								L	C	S	
Ref. [21]	Boost	Non pulsating	$\frac{1}{1-d}$	d	$0 < d < 1$	U_{high}	$\frac{I_{high}}{1-d}$	1	1	2	Yes
Ref. [68]	Interleaved	Low ripple	$\frac{3}{1-d}$	$\frac{d}{3}$	$0.67 < d < 1$	$\frac{2}{3}U_{high}$	$\frac{2I_{high}}{1-d}$	3	3	6	No
Ref. [30]	Interleaved + Three level	Low ripple	$\frac{2}{1-d}$	$\frac{d}{2}$	$0 < d < 0.5$ $0.5 < d < 1$	$\frac{U_{high}}{2}$	$\frac{I_{high}}{d(1-d)}$	2	4	5	No
Ref. [31]	Three level	Non pulsating	$\frac{2}{1-d}$	$\frac{d}{2}$	$0 < d < 1$	$\frac{U_{high}}{2}$	$\frac{2I_{high}}{1-d}$	1	3	4	No
Ref. [36]	VM	Non pulsating	$\frac{2}{1-d}$	$\frac{d}{2}$	$0 < d < 1$	$\frac{U_{high}}{2}$	$\frac{(1+d)I_{high}}{d(1-d)}$	1	4	4	Yes
Ref. [9]	Quasi-Z source	Non pulsating	$\frac{1+d}{1-d}$	$\frac{d}{2-d}$	$0 < d < 1$	$\frac{U_{high}}{1+d}$	$\frac{2I_{high}}{1-d}$	2	4	3	Yes
Ref. [41]	H bridge	Non pulsating	$\frac{1}{1-2d}$	$\frac{2d}{2d-1}$	Up: $0 < d < 0.5$ Down: $0.5 < d < 1$	U_{high}	$\frac{dI_{high}}{(1-d)(1-2d)}$	1	3	5	Yes
Ref. [10]	Cascaded	Non pulsating	$\frac{2+d}{1-d}$	$\frac{d}{3-d}$	$0 < d < 1$	$\frac{U_{high}}{2+d}$	$\frac{(1+2d)I_{high}}{d(1-d)}$	2	6	5	No
Ref. [46]	Cascaded	pulsating	$\frac{1+d-d^2}{(1-d)^2}$	$\frac{d^2}{1+d-d^2}$	$0 < d < 1$	$\frac{U_{high}}{1+d-d^2}$	$\frac{(1-d)^2 I_{high}}{1+d-d^2}$	2	4	5	No

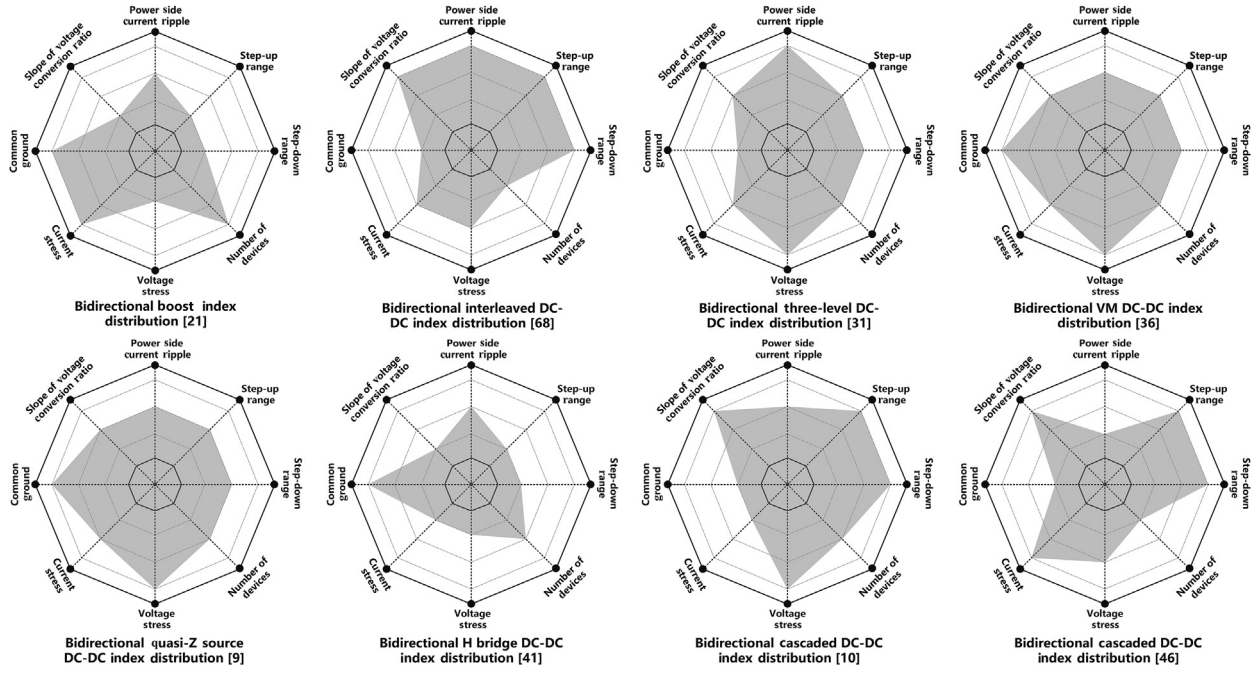


Fig. 27. Distribution of indexes of typical non-isolated bidirectional DC-DC topologies.

In Fig. 27, a larger shaded region indicates better indicators. From the inside to the outside, there are three grades: poor, medium, and excellent, respectively. Owing to the effect of parasitic parameters, for a very high duty cycle, the power loss increases and the efficiency of the converter decreases [18]. Thus, the duty cycle generally does not exceed 0.8 in practical engineering. Since it takes some time to turn on and turn off the switches, the minimum duty cycle is generally not less than 0.2 in practical engineering. The voltage conversion ratio function cannot be directly compared, however, the value of $d = 0.8$ reflects the width of the voltage conversion range. The stress function cannot be compared directly, however, the value of $d = 0.8$ reflects the maximum voltage and current stresses that the devices may withstand. The slope of the voltage conversion ratio is the value corresponding to $d = 0.8$, which reflects the maximum slope of voltage conversion ratio that may appear.

3.4. Comparison and evaluation of typical isolated bidirectional DC-DC converter

This section compares typical isolated bidirectional DC-DC topologies from six aspects: power source side current ripple, voltage and

current stresses, power density, number of devices, and transformer winding design. The distribution of indexes for seven typical isolated bidirectional DC-DC topologies are summarized in Table 5.

This section evaluates seven typical isolated bidirectional DC-DC converter topologies on six aspects: power side current ripple, voltage stress, current stress, power density, numbers of devices, and transformer winding design, as shown in Fig. 28.

4. Limitations and prospects of bidirectional DC-DC converters for HESS

Herein, the research status of bidirectional DC-DC converter topologies are summarized and compared, and the future research directions of bidirectional DC-DC for HESS are prospected, aiming to further promote the development of NEV and help the use of green energy and carbon reduction. Topologies of bidirectional DC-DC conversion systems for vehicle powertrain have become a research hotspot with the development of NEV. On the basis of bidirectional DC-DC topology optimization, designing an excellent DC-DC conversion system while meeting the requirements for HESS is significantly challenging. The main issues about bidirectional DC-DC conversion systems for NEV powertrain are as follows:

Table 5
Indexes of typical isolated bidirectional DC-DC topologies.

Reference	Technology used	Power source side current ripple	Voltage stress of switches	Transformer design	Current stress of switches	Number of devices				Power density
						L	T	C	S	
Ref. [48]	Flyback	Pulsating	U_{high}	Difficulty	$\frac{I_{high}}{1-d}$	0	1	2	2	Low
Ref. [51]	Resonant DAB	Pulsating	$\frac{U_{high}}{2}$	Medium	I_{high}	2	1	4	8	Middle
Ref. [55]	Non-resonant DAB	Non pulsating	$\frac{U_{high}}{2}$	Easy	I_{high}	1	1	2	8	Middle
Ref. [57]	Z source full bridge	Pulsating	U_{high}	Easy	$\frac{2I_{high}}{1-2d}$	4	1	4	12	Middle
Ref. [59]	Quasi-Z source full bridge	Non pulsating	U_{high}	Easy	$\frac{2I_{high}}{1-d}$	4	1	4	12	Middle
Ref. [60]	Two-phase interleaved	Low ripple	$\frac{U_{high}}{2}$	Medium	$\frac{I_{high}}{4}$	3	2	1	16	Low
Ref. [63]	Three phase Y-Y interleaved	Low ripple	$\frac{U_{high}}{2}$	Difficulty	$\frac{I_{high}}{3}$	3	3	5	12	High

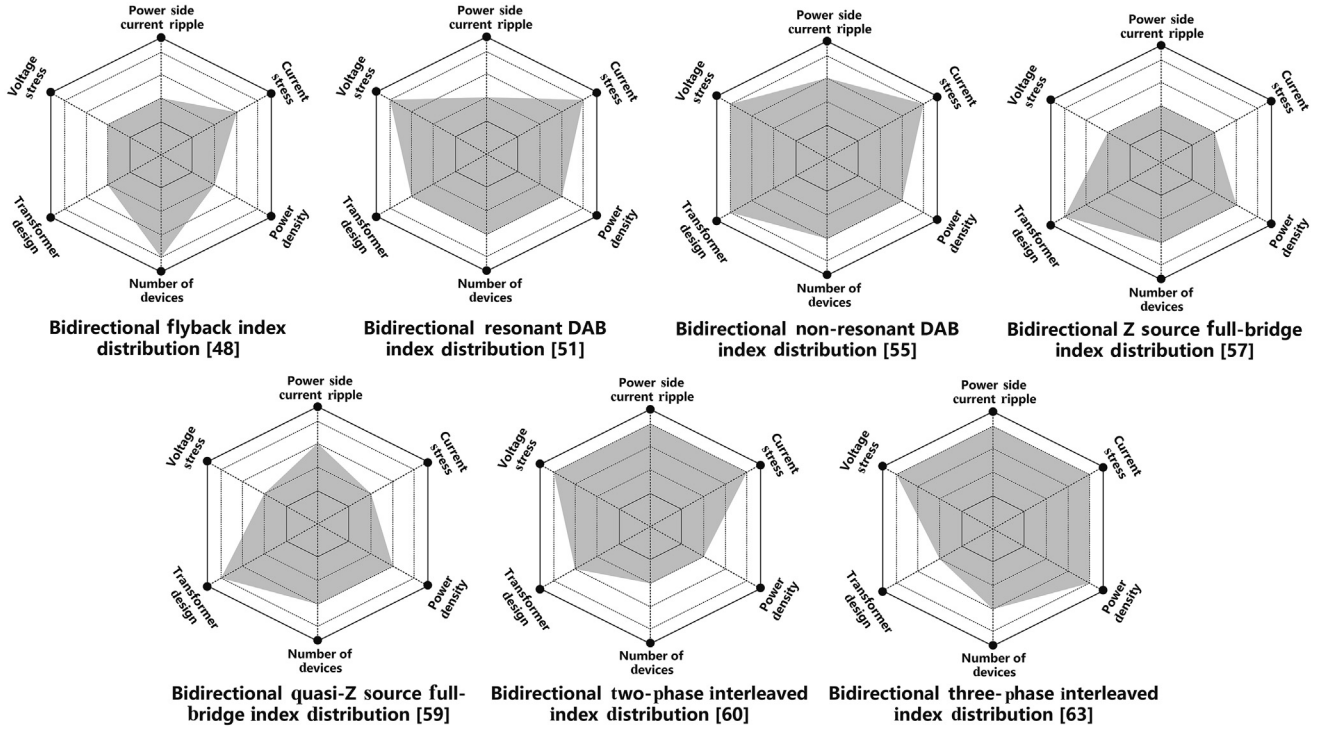


Fig. 28. Distribution of indexes of typical isolated bidirectional DC-DC topologies.

With continuously improved bus voltage levels (400 V promoted to 800 V [69]) of powertrain, a bidirectional DC-DC converter is required to continuously improve the voltage conversion ratio to match the SC (or power battery) and vehicle bus voltages. With continuously improved energy efficiency requirements of the vehicle, stricter requirements are put forward for the conversion efficiency of bidirectional DC-DC converter. Therefore, optimizing topologies of converters are complicated considering the system loss and stresses of devices.

The instantaneous power required by the NEV will change in real time with the change of driving conditions, resulting in the fluctuation of bidirectional DC-DC demand voltage and power in a wide range. Therefore, the bidirectional DC-DC converter can output stable voltage under all operating conditions, while reducing the power source side current ripple. It is important to address issues pertaining to topology design, accurate modeling, and optimal control of the bidirectional converters for HESS that need to be solved urgently.

A bidirectional DC-DC converter is a strong nonlinear system. Under specific conditions, there would exist various types of bifurcation, chaos, and other nonlinear phenomena. These are embodied in the increased voltage and current ripples of the converter, the increased harmonics, the decreased conversion efficiency, noise, oscillations, and even system collapse. The procedure for performing nonlinear modeling of DC-DC converters, analyzing converter stability, and implementing chaos control, constitute important research topics related to bidirectional DC-DC converters.

Using devices with wideband gaps (SiC, GaN) can result in the converter operating with a high switching frequency, significantly reducing the volume of energy storage elements in a converter. This is an important means to improve the power density. However, when the switching frequency of a converter is too high, the EMI would become severe. Balancing the relationship between the switching frequency and the electromagnetic compatibility of a system in the application of devices with wideband gaps would also be an important direction in the research on topology design and the modeling of bidirectional DC-DC converters for HESS.

The reliability of a bidirectional DC-DC converter for vehicle powertrain is an important factor that seriously affects the development of the NEV industry. Completely combining the researches of system detection, modeling, control strategy, and fault diagnosis methods to realize the safe and reliable operation of bidirectional DC-DC conversion systems would also be an important research direction regarding bidirectional converters for HESS.

Considering the above mentioned problems faced during research on bidirectional DC-DC converters for HESS, related studies on the following aspects can be considered.

With regards to the issue of high voltage conversion ratio and efficiency operation of the bidirectional DC-DC converters for HESS, the advantages of devices with wideband gaps, such as high frequency and low loss, can be utilized. The optimization calculation method can be used to improve and optimize bidirectional DC-DC converters based on existing interleaved, quasi-Z source, cascaded, and other topologies. This can be done in terms of decreasing the volume of energy storage elements, reducing the stress of devices, and reducing switching loss.

To solve the issue of stable operation under all operating conditions and a low power side current ripple of the bidirectional DC-DC converter for HESS, the dynamic response and anti-interference ability of the converter can be synthetically improved based on existing topology optimization, such as implementing an interleaved parallel technology, combined with research on accurate modeling and the adaptive control strategy.

To solve the issue of nonlinear modeling and control of the bidirectional DC-DC converter for HESS, based on the study of nonlinear dynamics and combined synthetical modeling methods, such as state variable equation and discrete mapping, the nonlinear behavior caused by changes in the bidirectional DC-DC converter parameter can be revealed by qualitative and quantitative methods. This is done to apply chaos control to the converter and realize the expected periodicity.

Considering the issues regarding the application of wideband gaps semiconductor devices in bidirectional DC-DC converters, the

electromagnetic compatibility characteristic of the system in different frequency domains can be analyzed, and accurate EMI models can be subsequently built on the basis of EMI modeling.

To improve the reliability of the bidirectional DC–DC converter, the typical dynamic behavior of the converter under different fault conditions, the fault behavior laws under different control strategies, the fault characteristics and influencing factors, and the quantitative extraction of key fault feature of fault external characteristics in each stage are analyzed. This is to realize the construction of a fault model of the bidirectional DC–DC converter, providing a model basis for the fault diagnosis of bidirectional DC–DC converters.

5. Conclusion

This paper systematically summarizes the research status of bidirectional DC–DC topologies for HESS. The related researches in recent years are summarized and evaluated from two aspects: non-isolated and isolated bidirectional DC–DC topologies.

Based on the introduction of the structure and characteristics of HESS for NEV powertrain, the research status of bidirectional DC–DC converters for HESS are analyzed. The non-isolated and isolated bidirectional DC–DC topologies for HESS are comprehensively reviewed. The advantages and disadvantages of these topologies are expounded, and the optimization directions for HESS are pointed out.

On this basis, the evaluation system of the HESS is established, and the non-isolated and isolated topologies are compared and evaluated from eight and six dimensions, respectively.

Lastly, to further promote the development of NEV and help the use of green energy and carbon reduction, the issues regarding bidirectional DC–DC topologies optimization of HESS are summarized and the future research directions are prospected.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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